

Technical Note
**Prepaid Accelerated Share Repurchase (ASR): Average-Price Risk,
VaR/ES, and Implementation Under TWAP/VWAP and ADTV
Constraints**

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Abstract

A prepaid Accelerated Share Repurchase (ASR) transfers to the dealer an initial short stock position created by the up-front share delivery; the dealer then covers that short over the execution horizon. The core market risk is the randomness of the *average execution price*—often proxied by a time-weighted (TWAP) or volume-weighted (market-VWAP) average—whose distribution depends on both price dynamics and execution constraints (e.g., participation caps such as 10% ADTV). This note develops: (i) the theoretical covariance structure of geometric Brownian motion (GBM), (ii) the exact variance of the arithmetic average price, (iii) the resulting prepaid-ASR cover-cost VaR/ES mapping in normalized units (as a fraction of prepaid P), and (iv) implementation scenarios spanning unconstrained TWAP, market-VWAP, and ADTV-constrained execution with binding/non-binding regimes.

1 Executive Summary

A prepaid ASR exposes the dealer to the right tail of the *cost of covering* the up-front delivered shares. Normalizing by the prepaid amount P reduces the problem to the distribution of the dealer’s average execution price relative to S_0 . Under a GBM price model, the key object is the covariance kernel

$$\mathbb{E}[S_t S_u] = S_0^2 \exp(\mu(t+u) + \sigma^2 \min(t, u)),$$

which yields an exact closed form for $\text{Var}(A_{\text{TWAP}})$ and an exact conditional form for $\text{Var}(A_{\text{VWAP}} | V)$. VaR/ES require a distributional step; a practical choice is moment-matched lognormal for the normalized average price.

ADTV constraints enter through the diagnostic

$$\eta_{\text{liq}} = \frac{Q}{\rho \cdot \text{ADTV}},$$

where Q is the required share quantity to repurchase and ρ is the participation rate (e.g., 10%). If $\eta_{\text{liq}} \leq T$, the constraint is non-binding and VWAP variance reduction is achievable; if $\eta_{\text{liq}} > T$, the constraint is binding and VWAP behaves closer to TWAP (and may effectively “stretch” the averaging horizon).

2 Contract and Risk Variable Definition

2.1 Contract primitives and notation

We use the following notation:

- P : prepaid amount (cash received by dealer at inception),
- f : up-front delivery fraction (e.g., $f = 0.8$),
- S_0 : spot or reference price at inception,
- T : execution horizon (e.g., 6 months),
- N_0 : initial shares delivered to client,
- Q : shares the dealer must buy (cover) to neutralize initial short,
- $\bar{S}$: realized average execution price paid by dealer,
- C : cover cost $= Q\bar{S}$,
- α : confidence level (e.g., 98%),
- $\text{VaR}_\alpha, \text{ES}_\alpha$: Value-at-Risk and Expected Shortfall at level α .

2.2 Up-front shares and cover cost

A common approximation for up-front delivered shares is

$$N_0 \approx f \frac{P}{S_0}. \quad (1)$$

Focusing on the market risk of covering the initial short, we take $Q = N_0$ and define cover cost

$$C = N_0 \bar{S}. \quad (2)$$

2.3 Normalized cover cost and “exposure beyond prepaid”

Normalize by P :

$$\boxed{\frac{C}{P} = f \frac{\bar{S}}{S_0}}. \quad (3)$$

Because P is received up-front, a natural “loss beyond prepaid” is

$$L = (C - P)^+, \quad \frac{L}{P} = \left(\frac{C}{P} - 1 \right)^+. \quad (4)$$

Thus, at confidence level α ,

$$\boxed{\text{VaR}_\alpha(L) = P \cdot \max(\text{VaR}_\alpha(C/P) - 1, 0)}. \quad (5)$$

This expresses the observation that meaningful exposure only occurs when the tail of C/P crosses 1.

3 Price Model and Covariance Structure

3.1 GBM dynamics

Assume geometric Brownian motion (GBM):

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t, \quad S_0 > 0. \quad (6)$$

The explicit solution is

$$S_t = S_0 \exp\left((\mu - \tfrac{1}{2}\sigma^2)t + \sigma W_t\right). \quad (7)$$

3.2 Exact cross moment

For all $t, u \geq 0$,

$$\mathbb{E}[S_t S_u] = S_0^2 \exp(\mu(t+u) + \sigma^2 \min(t, u)). \quad (8)$$

This identity drives average-price variance computations via double integration.

4 TWAP Implementation: Exact Variance of the Arithmetic Average

4.1 Definition (TWAP proxy)

Define the arithmetic (time) average price:

$$A_{\text{TWAP}} = \frac{1}{T} \int_0^T S_t dt. \quad (9)$$

4.2 Exact mean

$$\mathbb{E}[A_{\text{TWAP}}] = \frac{S_0}{T} \int_0^T e^{\mu t} dt = S_0 \frac{e^{\mu T} - 1}{\mu T} \quad (\mu \neq 0), \quad (10)$$

with continuous limit $\mathbb{E}[A_{\text{TWAP}}] = S_0$ as $\mu \rightarrow 0$.

4.3 Exact second moment and variance (no approximation)

Start from

$$A_{\text{TWAP}}^2 = \frac{1}{T^2} \int_0^T \int_0^T S_t S_u dt du. \quad (11)$$

Taking expectations and using symmetry of $\min(t, u)$, one obtains

$$\mathbb{E}[A_{\text{TWAP}}^2] = \frac{2S_0^2}{T^2(\mu + \sigma^2)} \left[\frac{e^{(2\mu + \sigma^2)T} - 1}{2\mu + \sigma^2} - \frac{e^{\mu T} - 1}{\mu} \right]. \quad (12)$$

Hence

$$\text{Var}(A_{\text{TWAP}}) = \mathbb{E}[A_{\text{TWAP}}^2] - \left(S_0 \frac{e^{\mu T} - 1}{\mu T} \right)^2. \quad (13)$$

Risk-measure specialization ($\mu = 0$), still exact. Taking $\mu \rightarrow 0$ gives

$$\text{Var}(A_{\text{TWAP}}) = \frac{2S_0^2}{T^2\sigma^2} \left[\frac{e^{\sigma^2 T} - 1}{\sigma^2} - T \right] - S_0^2. \quad (14)$$

Normalized:

$$V_{\text{TWAP}} := \text{Var}\left(\frac{A_{\text{TWAP}}}{S_0}\right) = \frac{2}{T^2\sigma^2} \left[\frac{e^{\sigma^2 T} - 1}{\sigma^2} - T \right] - 1. \quad (15)$$

5 Market-VWAP Implementation: Exact Conditional Variance

5.1 Definition (market VWAP)

Let V_t be cumulative market volume (shares). Market-VWAP is

$$A_{\text{VWAP}} = \frac{\int_0^T S_t dV_t}{V_T}. \quad (16)$$

Define the weight measure

$$w(t) := \frac{dV_t}{V_T}, \quad \int_0^T w(t) = 1, \quad (17)$$

so

$$A_{\text{VWAP}} = \int_0^T S_t w(t). \quad (18)$$

5.2 Exact conditional variance (given a realized volume path)

Condition on V (so w is deterministic). For $\mu = 0$,

$$\boxed{\text{Var}(A_{\text{VWAP}} \mid V) = S_0^2 \int_0^T \int_0^T w(t)w(u) \left(e^{\sigma^2 \min(t,u)} - 1 \right) dt du.} \quad (19)$$

This is the VWAP analogue of the TWAP expression: same kernel, different weights.

5.3 Unconditional variance requires a volume model

By the law of total variance,

$$\text{Var}(A_{\text{VWAP}}) = \mathbb{E}[\text{Var}(A_{\text{VWAP}} \mid V)] + \text{Var}(\mathbb{E}[A_{\text{VWAP}} \mid V]). \quad (20)$$

Under the baseline assumption that volume is independent of price and $\mu = 0$, one has $\mathbb{E}[A_{\text{VWAP}} \mid V] = S_0$ and the second term vanishes. One then computes the first term by specifying either (i) a deterministic expected volume curve $w(t)$, or (ii) a stochastic volume model for V_t and averaging the conditional variance.

6 Mapping Average-Price Variance to ASR VaR/ES

6.1 Variance of normalized cover cost

Using $\frac{C}{P} = f\left(\frac{\bar{S}}{S_0}\right)$,

$$\boxed{\text{Var}\left(\frac{C}{P}\right) = f^2 \text{Var}\left(\frac{\bar{S}}{S_0}\right).} \quad (21)$$

Choose $\bar{S}$ as A_{TWAP} , A_{VWAP} , or the realized average paid under a simulated constrained execution policy.

6.2 Distributional step (required for VaR/ES)

Variance alone does not uniquely determine VaR/ES. A standard practical approach is to approximate

$$Y := \frac{\bar{S}}{S_0} \sim \text{LogNormal}(m, s^2) \quad (22)$$

by moment matching:

$$\mathbb{E}[Y] = 1, \quad \text{Var}(Y) = V. \quad (23)$$

Then

$$\boxed{s^2 = \ln(1 + V), \quad m = -\frac{1}{2}s^2.} \quad (24)$$

6.3 Closed-form VaR and ES under lognormal matching

Let z_α denote the standard normal α -quantile and Φ the standard normal CDF.

VaR of normalized cover cost.

$$\boxed{\text{VaR}_\alpha\left(\frac{C}{P}\right) = f \cdot \exp\left(-\frac{1}{2}s^2 + sz_\alpha\right), \quad s^2 = \ln(1 + V).} \quad (25)$$

ES of normalized cover cost. For $Y \sim \text{LogNormal}(m, s^2)$,

$$\text{ES}_\alpha(Y) = \frac{\exp(m + \frac{1}{2}s^2) \Phi(s - z_\alpha)}{1 - \alpha}. \quad (26)$$

Under the moment-matching $\exp(m + \frac{1}{2}s^2) = \mathbb{E}[Y] = 1$, hence

$$\boxed{\text{ES}_\alpha\left(\frac{C}{P}\right) = f \cdot \frac{\Phi(s - z_\alpha)}{1 - \alpha}.} \quad (27)$$

VaR of loss beyond prepaid.

$$\boxed{\text{VaR}_\alpha(L) = P \cdot \max(\text{VaR}_\alpha(C/P) - 1, 0).} \quad (28)$$

7 Liquidity / ADTV (Participation) Constraint: 10% ADTV

7.1 Definition and binding test

Let ADTV be average daily traded volume (shares/day). A participation cap $\rho = 10\%$ imposes

$$\text{daily buy} \leq \rho \cdot \text{ADTV}. \quad (29)$$

If Q shares must be bought, the implied minimum execution time is

$$\boxed{\eta_{\text{liq}} = \frac{Q}{\rho \cdot \text{ADTV}}.} \quad (30)$$

The regime is determined by

$$\eta_{\text{liq}} \leq T \Rightarrow \text{non-binding}, \quad \eta_{\text{liq}} > T \Rightarrow \text{binding}. \quad (31)$$

7.2 Practical variance adjustment logic

In a transparent risk implementation, treat the binding case as a constraint-inflated average-price variance problem:

- **Non-binding:** use market-VWAP variance based on the selected $w(t)$ (Section 5).
- **Binding:** revert toward TWAP variance (conservative) or inflate VWAP variance by a factor linked to η_{liq}/T ; alternatively, run constrained execution simulation with $u_t \leq \rho V_t$ and compute empirical VaR/ES.

8 Implementation Scenarios (End-to-End Recipes)

Scenario A: TWAP, unconstrained (transparent / conservative)

1. Set $\bar{S} = A_{\text{TWAP}}$.
2. Compute $V_{\text{TWAP}} = \text{Var}(A_{\text{TWAP}}/S_0)$ from the exact formula (Section 4).
3. Map to $\text{VaR}_\alpha(C/P)$ and $\text{ES}_\alpha(C/P)$ via lognormal matching; scale by f .
4. Compute exposure beyond prepaid via $\max(\text{VaR}_\alpha(C/P) - 1, 0)$.

Scenario B: Market-VWAP, unconstrained (contract-consistent)

1. Choose a deterministic $w(t)$ (expected volume curve) or a stochastic volume model.
2. Compute $\text{Var}(A_{\text{VWAP}} | V)$ via the weighted double integral (Section 5).
3. Average over volume if stochastic; otherwise use deterministic $w(t)$.
4. Map to $\text{VaR}_\alpha(C/P)$ and $\text{ES}_\alpha(C/P)$ via lognormal matching; scale by f .

Scenario C: 10% ADTV, non-binding (VWAP-like execution feasible)

1. Compute $Q = fP/S_0$ (or use actual N_0).
2. Compute $\eta_{\text{liq}} = Q/(0.10 \cdot \text{ADTV})$.
3. If $\eta_{\text{liq}} \leq T$, proceed with Scenario B.

Scenario D: 10% ADTV, binding (liquidity dominates)

1. Compute η_{liq} and confirm $\eta_{\text{liq}} > T$.
2. Conservative: use Scenario A (TWAP variance) as an upper bound.
3. More nuanced: inflate VWAP variance as a function of η_{liq}/T , or run constrained execution simulation.

Scenario E: Full execution + impact (desk-true)

1. Specify an S_t model (GBM / local vol / stochastic vol).
2. Specify market volume V_t and participation cap $u_t \leq \rho V_t$.
3. Optionally include temporary impact $\tilde{S}_t = S_t + \kappa u_t$ and spread.
4. Simulate execution policy (TWAP/VWAP tracking/adaptive) and compute $C^{(i)}$ path-wise.
5. Compute empirical VaR/ES of C/P and $(C/P - 1)^+$.

9 Model Risk and Practical Considerations

- **Measure choice:** VaR/ES should be computed under a real-world risk measure; setting $\mu = 0$ is common for medium-horizon market risk.
- **Volume/price dependence:** volume is not strictly independent of returns; correlations can shift VWAP variance and tail risk.
- **Execution adaptivity:** adaptive strategies reduce variance relative to passive TWAP/VWAP but are limited by caps, risk limits, and impact.
- **Borrow and recall risk:** stock borrow costs and recalls can dominate tail outcomes and should be layered as separate stress factors where material.

A Formula Block

A1. Exact TWAP normalized variance ($\mu = 0$)

$$V_{\text{TWAP}} = \text{Var}\left(\frac{A_{\text{TWAP}}}{S_0}\right) = \frac{2}{T^2\sigma^2} \left[\frac{e^{\sigma^2 T} - 1}{\sigma^2} - T \right] - 1.$$

A2. VWAP conditional variance ($\mu = 0$)

$$\text{Var}\left(\frac{A_{\text{VWAP}}}{S_0} \middle| V\right) = \int_0^T \int_0^T w(t)w(u) \left(e^{\sigma^2 \min(t,u)} - 1 \right) dt du, \quad w(t) = \frac{dV_t}{V_T}.$$

A3. Lognormal mapping ($\mathbb{E}[Y] = 1$, $\text{Var}(Y) = V$)

$$s^2 = \ln(1 + V), \quad m = -\frac{1}{2}s^2.$$

A4. VaR/ES of normalized cover cost

$$\text{VaR}_\alpha(C/P) = f \exp(-\frac{1}{2}s^2 + sz_\alpha), \quad \text{ES}_\alpha(C/P) = f \frac{\Phi(s - z_\alpha)}{1 - \alpha}.$$

A5. Exposure beyond prepaid

$$\text{VaR}_\alpha(L) = P \cdot \max(\text{VaR}_\alpha(C/P) - 1, 0).$$

A6. 10% ADTV binding test

$$\eta_{\text{liq}} = \frac{Q}{0.10 \cdot \text{ADTV}} \Rightarrow \begin{cases} \eta_{\text{liq}} \leq T & \text{non-binding,} \\ \eta_{\text{liq}} > T & \text{binding.} \end{cases}$$

B What is $\bar{S}$?

- **TWAP proxy:** $\bar{S} = A_{\text{TWAP}}$.
- **Market-VWAP proxy:** $\bar{S} = A_{\text{VWAP}}$.
- **Execution simulation:** $\bar{S} = C/Q$ (realized average paid, including impact if modeled).